

ECC - Problems

1) The negative of point $p = (x_p, y_p)$ is given by

$$\boxed{-p = (x_p, -y_p \cdot q)}$$

Prob1:

What are the negatives of the following elliptic curves over $\mathbb{Z}_7$? $P = (5, 8)$; $Q = (3, 0)$; $R = (0, 6)$

Soln:

$$P = (5, 8) \Rightarrow x_p = 5 \text{ \& } y_p = 8, q = 17 \text{ (prime)}$$

$$-P = (5, -8 \cdot 17) = (5, -8) = (5, 9)$$

$\xrightarrow{\text{positive } -8+17=9}$

$$\boxed{-P = (5, 9)}$$

$$-Q = (3, -0 \cdot 17) = (3, 0) \quad \boxed{Q = (3, 0)}$$

$$-R = (0, -6 \cdot 17) = (0, -6) = (0, 11)$$

$$\boxed{-R = (0, 11)}$$

Prob2:

For $E_{11}(1, 6)$, Consider the point $G = (2, 7)$. Compute the multiples of G from $2G$ through $13G$.

Soln:

We follow the rules of addition.

To Compute $2G$: $\underline{(2, 7) + (2, 7)}$

$$\begin{aligned} \lambda &= (3x_p^2 + a/2y_p) \bmod P \\ &= (3(2)^2 + 1/2(7)) \cdot 11 \\ &= 12 + 1/14 \cdot 11 = \frac{13}{14} \cdot 11 \end{aligned}$$

→ similar points $\underline{(G+G)}$

(here $x_p = 2$; $y_p = 7$)

$\phi E_p(a, b) = E_{11}(1, 6)$

$\underline{\frac{P=11}{a=1; b=6}}$

$$14\lambda = 13 \cdot 11 = 13 \quad \text{--- (1)}$$

Apply 7.11 on both sides

(2)

$$\cancel{\lambda = 14} \cdot \cancel{11} = 14 \cdot \lambda \cdot 11 = 13 \cdot 11$$

When $\lambda = 1$

$$14 \cdot 11 \neq 13 \cdot 11$$

$\lambda = 2$

$$28 \cdot 11 \neq 13 \cdot 11$$

$\lambda = 3$

$$42 \cdot 11 \neq 13 \cdot 11$$

$\vdots$

$\lambda = 8$

$$112 \cdot 11 = 13 \cdot 11$$

$$2 = 2$$

$$\boxed{\lambda = 8}$$

New Coordinates $(X_R, Y_R) \Rightarrow 2G$

$$X_R = (\lambda^2 - x_P - x_Q) \cdot p$$

$$= (8^2 - 2 - 2) \cdot 11$$

$$\boxed{X_R = 5}$$

$$Y_R = (\lambda (x_P - x_R) - y_P) \cdot p$$

$$= (8(2-5) - 7) \cdot 11 = -9 \text{ [or] } 2$$

$$\boxed{2G = (5, 2)}$$

Similarly $3G = 2G + G$ i.e. $(x_P, y_P) + (x_Q, y_Q)$

$$\lambda = \left(\frac{y_Q - y_P}{x_Q - x_P} \right) \cdot p + Q$$

$$= \frac{7-2}{2-5} \cdot 11 = 0$$

$\lambda = \frac{y_Q - y_P}{x_Q - x_P} \text{ mod } p \Rightarrow \text{diff. point}$
 $\underline{G + 2G}$

$$= \frac{7-2}{2-5} \cdot 11 = \frac{5}{-3} \cdot 11$$

$$\boxed{\lambda = 2}$$

$$\frac{11}{112}$$

$$110$$

$$\frac{3}{114}$$

New Coordinates $(X_R, Y_R) \Rightarrow 3G$

$$\begin{aligned} X_R &= (\lambda^2 - x_P - x_Q) \cdot P \\ &= (2^2 - 5 - 2) \cdot 11 \\ \boxed{X_R} &= \boxed{8} \end{aligned} \quad \left| \quad \begin{aligned} Y_R &= (\lambda (x_P - x_Q) - y_P) \cdot P \\ &= (2(5 - 2) - 2) \cdot 11 \\ &= (-8 \cdot 11) = 3 \\ \boxed{Y_R} &= \boxed{3} \end{aligned} \right.$$

$$3G = (8, 3)$$

$$4G = 2G + 2G \quad \text{i.e. } P = (5, 2) + Q = (5, 2) \Rightarrow P + Q = R$$

$2G + 2G = 4G$

$$\begin{aligned} \lambda &= (3x_P^2 + a/2y_P) \cdot P \\ &= 3(5)^2 + 1/2(2) \cdot 11 = 76/4 \cdot 11 = 19 \cdot 11 = \underline{8} \end{aligned}$$

4G Coordinates (x_P, y_R) :

$$\begin{aligned} X_R &= (\lambda^2 - x_P - x_Q) \cdot P \\ &= (8^2 - 5 - 5) \cdot 11 \\ \boxed{X_R} &= \boxed{10} \end{aligned} \quad \left| \quad \begin{aligned} Y_R &= (\lambda (x_P - x_Q) - y_P) \cdot P \\ &= (8(5 - 10) - 2) \cdot 11 \\ \boxed{Y_R} &= \boxed{2} \end{aligned} \right.$$

$$\boxed{4G = (10, 2)}$$

$$5G = 3G + 2G \Rightarrow (8, 3) + (5, 2)$$

$$\lambda = \frac{y_Q - y_P}{x_Q - x_P} \pmod{P}$$

$$\lambda = \frac{2 - 3}{5 - 8} \cdot 11 = +10 \cdot 8^{-1} \cdot 11 = 10 \cdot 7 \cdot 11 = 4$$

$$X_R = (4^2 - 8 - 5) \cdot 11 = 3 \cdot 11 = 3$$

$$Y_R = (4(8 - 3) - 3) \cdot 11 = 17 \cdot 11 = 6$$

$$\boxed{5G = (4, 6)}$$

$$6G = 3G + 3G = (7, 9)$$

$$7G = 3G + 4G = (7, 2)$$

$$8G = 4G + 4G = (3, 5)$$

$$9G = 5G + 4G = (10, 2)$$

$$10G = 5G + 5G = (8, 8)$$

$$11G = 5G + 6G = (5, 9)$$

$$12G = 6G + 6G = (8, 8)$$

$$13G = 9G + 4G = (3, 5)$$

(4)

3) Perform EC encryption/decryption using the cryptosystem parameters $E_1(1,6)$ and $G = (2,7)$. B's secret key is $n_B = 7$

a. Find B's public key, P_B .

b. 'A' wishes to encrypt the message $P_m = (10,9)$ and chooses the random ^{value} ~~variable~~, $k = 3$. Determine Ciphertext C_m .

c. Show the Calculation by which B recovers P_m from C_m .

Soln:

a) $P_B = n_B \times G$

$$= 7 \times (2,7) = 7G$$

$$P_B = (7,2)$$

$$7G = 3G + 4G$$

$$3G = 2G + G; 4G = 2G + 2G$$

$$7G = (7,2) \Rightarrow \text{Ref previous pblm.}$$

b) $C_m = \{kG, P_m + kP_B\}$

$$= \{3(2,7), (10,9) + 3(7,2)\}$$

$$= \{3G, (10,9) + (3,5)\} = \left\{ \frac{3G}{kG}, \frac{(10,2)}{P_m + kP_B} \right\}$$

c) $P_m = \frac{P_m + kP_B - n_B(kG)}{}$

$$= (10,2) - 7 \times (3G) = (10,2) - 7 \times (8,3)$$

$$= (10,2) - (3,5)$$

$$= (10,2) + (-3,5)$$

$$= (10,9)$$

$$-(3,5) = (3,-5) = (3,6) \text{ for } p=11$$

Ref URL: cse.sttx.ca/~ityang/csci-467/assignments/
CSCI467-assignment4-jason-anwar-plt